

Applying the Simplified Vocal Profile Analysis to Swiss German Dialects

Alessia Wüst¹, Adrian Leemann¹

¹ University of Berne, Switzerland

alessia.wuest@unibe.ch, adrian.leemann@unibe.ch

Abstract

This work presents the first perceptual investigation of vocal profiles in Swiss German, utilizing data from 127 speakers (aged 18-40) from the Swiss *Dialektatlas*. Using Simplified Vocal Profile Analysis, four coders assessed vocal profiles to map geographic and social trends. While inter-rater reliability proved challenging ($\kappa=0.09-0.39$), consensus coding revealed clear stratification. Regarding gender, men typically exhibited a relaxed vocal tract ('hypoarticulation'), whereas women showed a tendency toward pharyngeal expansion ('hyperarticulation'). Geographically, Northeast Switzerland is characterized by nasality, while Central Switzerland is marked by pharyngeal expansion – a finding likely confounded by the gender distribution in the sample. Notably, creaky voice remains rare in Swiss German. These results suggest that while vocal profiles function as salient markers of regional and social identity, methodological complexities call for refined protocols in future research.

Index Terms: Vocal Profile Analysis, Voice quality, Sociophonetics, Swiss German.

1. Introduction

While Swiss German segmental phonetics are extensively documented (e.g., [1, 2]), the suprasegmental domain of vocal profiles remains a significant research gap. Laver [3] and Abercrombie [4] define vocal profiles as the cumulative auditory coloring of the habitual vocal tract configuration. Despite flourishing research in Anglophone (e.g., [5, 6]) and Romance linguistics (e.g., [7, 8]), studies on German varieties remain sparse – and for Swiss German, virtually nonexistent beyond peripheral mentions in older dialect grammars. These historical accounts remained impressionistic, lacking a systematic coding framework to ensure reliability. They offer isolated observations rather than rigorous data on how voice quality is distributed across the population. This absence of evidence is critical, as in a dialect landscape as diverse as Switzerland's, a specific vocal setting likely acts as a salient 'badge' of identity – a function that has never been empirically tested. To do so, we present the first perceptual analysis of Swiss German vocal profiles using the Simplified Vocal Profile Analysis (SVPA) protocol [9]. Drawing on spontaneous speech from 127 speakers in the *Dialektatlas* corpus [1], this paper investigates the inter-rater reliability (IRR) of perceptual coding, explores gender-specific articulatory patterns, and creates the first geographic maps of vocal profile settings for German-speaking Switzerland.

1.1. Sociophonetic perspectives on voice quality

Pioneering work by Laver [3] established the Vocal Profile Analysis (VPA), distinguishing between anatomical constraints ('organic features') and culturally learned, habitual 'phonetic setting features' or long-term muscular biases. Research shows these settings are socially structured. Stuart-Smith's [5] work

on Glasgow English revealed strong correlations with class and gender, identifying working-class markers like open jaws, while associating men with creaky voice and women with whispery phonation. Beck and Schaeffler [10] found that among Scottish adolescents, variation was driven primarily by gender and biological maturation, with males actively lowering vocal tract resonances to project masculinity. Recently, White et al. [6] noted a rapid rise in creaky voice among young Australian women, contrasting with historical data where creak served as a 'marker of masculinity'. Cross-linguistic research reinforces these gendered norms. Pépiot and Arnold [7] demonstrated that female bilinguals exhibited higher breathiness than males across French and English, a pattern Mendoza et al. [11] similarly confirmed in Spanish.

1.2. Voice quality in German varieties

Research in the German-speaking context remains limited. Peters [12] investigated regional variation in the Bersenbrücker Land, finding that northern speakers exhibited 'harder' voices compared to 'softer, breathier' southern voices. He also noted that women generally displayed higher breathiness than men, ultimately concluding these qualities were markers of the speech community rather than the specific variety. While modern systematic studies of Swiss German are lacking, historical dialect grammars provide some impressionistic descriptions of vocal profiles, suggesting two primary zones: Northeast and Central Switzerland. In the Northeast, descriptions frequently center on nasality: Vetsch [13] described strong nasalization in Appenzell, and Wiget [14] observed a 'relaxed velum' in Toggenburg. However, this feature was not uniform; Berger [15] noted tense articulation in the St. Gallen Rheintal, while Streiff [16] and Weber [17] explicitly noted an absence of nasality in Glarus and Zürich, respectively. Central Swiss descriptions, by contrast, focus on specific lip and jaw settings. Schmid [18] highlighted strong labial activity and a small mouth opening in Entlebuch, whereas Clauß [19] noted weak lip activity and a small jaw angle in Uri.

1.3. Objectives

To move beyond impressionistic observations, this study subjects Swiss German vocal profiles to empirical scrutiny using the SVPA protocol developed by San Segundo and Mompean [9]. Applying this method to the *Dialektatlas* corpus [1] represents a novel methodological endeavor. This investigation is guided by three primary research questions:

1. **Reliability:** To what extent can the SVPA yield high IRR when applied to Swiss German data, and which specific settings (e.g., lip rounding) elicit the highest agreement among coders?
2. **Gender:** What specific vocal profile patterns – such as vocal tract tension or phonation type – can be identified as distinct gendered markers within the sample?

3. **Geography:** Are there empirically verifiable regional vocal profile settings that characterize specific Swiss dialect areas?

Based on previous literature and historical observations, we hypothesize that IRR will generally remain low (Fleiss' Kappa < 0.5). Regarding social variables, we anticipate that female speakers will exhibit greater breathiness, while males will tend toward creaky voice. Geographically, we expect to confirm the historical reports of a 'nasality belt' in Eastern Switzerland.

2. Methods

2.1. Speakers

Participants were selected from the Swiss *Dialäktatlas* corpus, a large-scale project documenting contemporary Swiss German of 1016 speakers across 127 survey sites. The full corpus features a stratified design with eight speakers per location: four older (60-80 years) and four younger (18-40 years), with two males and two females per age group. For the current perceptual investigation, a subsample of 127 speakers was established, representing one individual per analyzed location. To minimize age-related physiological changes – such as the natural reduction of vocal tract tension [20] – the sample was restricted to the younger cohort, excluding the older generation to avoid confounding variables. The resulting sample is gender-balanced, comprising 64 male and 63 female subjects. Due to the selection of a single speaker per survey point, the geographic distribution of gender is randomized, with each location represented by either a male or a female voice.

2.2. Materials

To ensure high-fidelity audio despite pandemic restrictions, the *Dialäktatlas* employed a hybrid remote strategy: While researchers supervised via Zoom, participants recorded speech locally using a smartphone application to bypass standard audio compression (recording quality of the *Dialäktatlas* corpus: 44.1kHz, 16 bit, cf. [21]). The current analysis exclusively used spontaneous speech derived from a ten-minute conversation conducted at the end of each of the interviews.

2.3. Data coding

For the perceptual analysis, we used the SVPA developed by San Segundo and Mompean [9]. Designed as an adaptation of Laver's [3] comprehensive scheme, the SVPA mitigates cognitive load and enhances replicability by reducing fine-grained scalar judgments to simplified categorical distinctions. Rather than rating degrees of deviation, raters coded for the presence of specific non-neutral settings (e.g., tense or lax) against a neutral baseline across ten vocal profile parameters. We adapted the protocol specifically for Swiss German phonology; for instance, the neutral baseline was defined to exclude vocalic nasalization. The coding team consisted of four raters with phonetic experience ranging from six months to over twenty years. Prior to data collection, the team underwent a rigorous calibration phase, initiated by a training session with San Segundo and followed by meetings in September 2025 to establish a reference sheet with auditory and visual anchors. Following the calibration phase, 127 spontaneous speech samples of about one minute each were assessed between October and November 2025. These individual judgments served as the basis for the calculations of IRR (IRR, see 2.4). Final decisions for the 127 profiles were determined during five synchronous

consensus review meetings where discrepancies were resolved through group re-evaluation.

2.4. Statistics

To evaluate the consistency of perceptual coding across the four independent raters, IRR was assessed using Fleiss' Kappa (κ). An unweighted Kappa coefficient was calculated individually for each of the ten vocal profile settings using the `kappam.fleiss()` function from the *irr* package in R [22]. Interpretation followed benchmarks established by Landis and Koch [23] and Kassambara [24]. According to these guidelines, $\kappa < 0$ indicates agreement below chance, while positive values are categorized as: 0.00–0.20 slight agreement, 0.21–0.40 fair agreement, and 0.41–0.60 moderate agreement. Statistical significance was assessed with a threshold of $p < 0.05$ to determine if the observed agreement exceeded random chance.

3. Results

3.1. Reliability

Fleiss' Kappa analysis revealed generally slight to fair reliability across the ten settings ($\kappa = 0.09$ – 0.386), though consensus significantly exceeded chance in all settings ($p < 0.05$). A distinct hierarchy emerged: four settings achieved fair agreement, led by velopharyngeal ($\kappa = 0.386$) and voice type ($\kappa = 0.302$). Conversely, five settings demonstrated only slight agreement ($\kappa < 0.20$), including laryngeal tension, laryngeal height, labial, mandibular, and pharyngeal, see Table 1.

Table 1: IRR (Fleiss' Kappa) and qualitative assessment of SVPA Settings

Rank	SVPA Setting	Kappa (κ)	z-Score	p-Value	Agreement Level
1	Velopharyngeal	0.386	10.7	< 0.05	Fair
2	Voice Type	0.302	10.8	< 0.05	Fair
3	Vocal Tract Tension	0.276	9.95	< 0.05	Fair
4	Apical	0.213	6.02	< 0.05	Fair
5	Laryngeal Tension	0.173	6.08	< 0.05	Slight
6	Laryngeal Height	0.161	5.96	< 0.05	Slight
7	Labial	0.126	4	< 0.05	Slight
8	Mandibular	0.095	3.29	< 0.05	Slight
9	Pharyngeal	0.094	2.85	< 0.05	Slight
–	Dorsal	NA	NA	NA	Undefined

'Dorsal' as a SVPA setting was always perceived as 'neutral', which is why it appears as 'NA' in Table 1.

3.2. Gender

The analysis of the 127 speakers (64 male, 63 female) revealed two distinct supralaryngeal patterns. While most speakers ($n = 77$) showed neutral vocal tract tension, a clear gender pattern emerged in this setting: men exhibited a relaxed vocal tract ($n = 24$) significantly more often than women ($n = 10$) ($\chi^2(2, N = 127) = 7.67, p = .022$), see Figure 1.

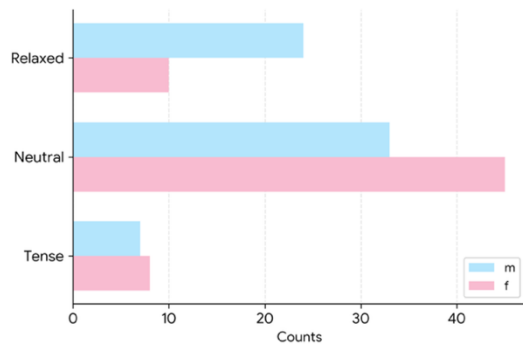


Figure 1: *Vocal tract tension by gender.*

Similarly, pharyngeal expansion appeared as a predominantly female characteristic (16 women vs. 3 men). This difference was statistically significant ($\chi^2(1, N=127) = 10.70, p=.001$). Regarding voice type, modal voice dominated ($n=112$); non-modal phonation was scarce, with creaky voice appearing in only 12 speakers (~9%; 7 men, 5 women). Other types were marginal (2 men harsh, 1 man breathy) or absent (whispered).

3.3. Geography

The perceptual analysis of the 127 speakers revealed clear geographic clusterings for specific voice quality settings, affirmatively answering the research question regarding the existence of regional voice quality markers. The mapping of the data was conducted using QGIS-LTR [25] with scalar interpolation (k nearest neighbor smoothing, $k=5$; cf. [26]). We found two primary regional phenomena: widespread nasality in the Northeast and pharyngeal expansion in Central Switzerland.

3.3.1. Velopharyngeal Setting (Nasality)

The most distinct regional pattern observed is the prevalence of nasality (velopharyngeal setting) in Eastern and Northeastern Switzerland. The data show a strong concentration of nasal profiles in the cantons of Appenzell, St. Gallen, Schaffhausen and Thurgau. While isolated instances of nasality were recorded in Central Switzerland and the Valais, the Northeast forms a coherent 'nasality belt', see Figure 2.

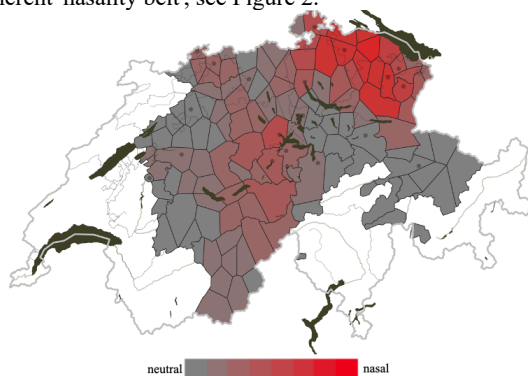


Figure 2: *Nasality by survey site (per-site consensus value, one speaker per site; kNN interpolation, $k=5$).*

3.3.2. Pharyngeal Setting (Expansion)

A second distinct regional marker is pharyngeal expansion, which characterizes the dialects of Central Switzerland. This feature is most prominent in the cantons of Uri and Schwyz, extending into Glarus and Graubünden. It is perceptually defined by hyperarticulation, creating the impression that the subject is 'speaking to an audience', see Figure 3.

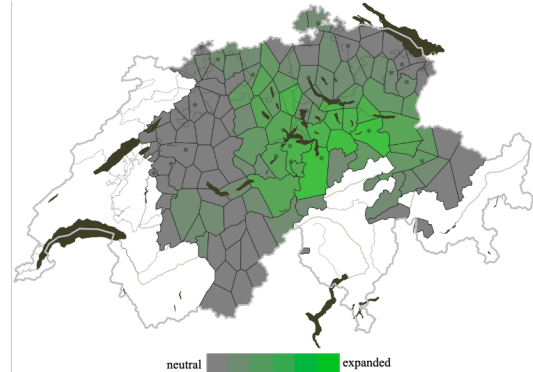


Figure 3: *Pharyngeal expansion by survey site (per-site consensus value, one speaker per site; kNN interpolation, $k=5$).*

In section 3.2 we reported that pharyngeal expansion varies significantly by gender, with women exhibiting higher rates. Keeping in mind that the gender distribution in our sample is geographically uneven (see 2.1), the question then emerges, whether the hotspots in Figure 3 may be an artifact of the gender effect. When we look at the raw gender distribution it turns out that females are slightly overrepresented in our sample in Central Swiss localities. Consequently, this gender bias coincides with – and may drive – the high degree of pharyngeal expansion observed in this Central Swiss region.

4. Discussion

4.1. Inter-Rater Reliability

IRR proved generally challenging, with Fleiss' Kappa values ranging from $\kappa=0.09$ to 0.39 . While falling within the fair to marginal range [23, 24] and trailing San Segundo and Mompean's [9] pilot results, these figures parallel Webb et al. [27], who reported similar ranges ($0.01-0.32$) using the original VPA. These results suggest that simplifying the protocol to binary decisions does not entirely eliminate the inherent subjectivity of auditory analysis. This perceptual instability underscores the inherent difficulty of applying discrete categories to a continuous acoustic signal. The fact that even experienced coders struggled with vocal tract settings suggests that these features may function as 'gestalts' – holistic impressions – rather than isolated articulatory maneuvers.

However, agreement levels were not random; rather, they appear to be structured by a combination of acoustic clarity and social salience. The highest reliability was observed for the velopharyngeal setting ($\kappa=0.386$, i.e., 'nasality'), likely because this feature is a highly recognizable marker of Eastern Swiss dialects. This supports the argument by Llamas et al. [28, p. 4] that certain linguistic forms are "key carriers of social meaning." In this case, because nasality is so socially prominent and 'marked' in the Swiss context – unlike in French, where it

is a standard part of the grammar [29] – it serves as a clear mental category for raters, leading to more consistent coding. Conversely, settings reliant on visual cues yielded the lowest agreement (mandibular $\kappa=0.09$; labial $\kappa=0.12$). Similarly, poor reliability in pharyngeal expansion ($\kappa=0.09$) highlights the ambiguity of impressionistic definitions like Beck's [30] 'strangled' quality, which is compounded by the acoustic difficulty of isolating pharyngeal constriction from laryngeal height [31].

4.2. Gender Distribution

The study found clear differences in men's and women's vocal profiles. Men were more likely to speak with a relaxed vocal tract ('hypoarticulation'; $n=24$ vs. $n=10$ women), while women significantly more often used pharyngeal expansion ('hyperarticulation'; $n=16$ vs. $n=3$ men). However, this finding must be interpreted with caution: the IRR for the pharyngeal setting was notably low ($\kappa=0.094$), indicating substantial disagreement among coders regarding the presence of this feature. Despite this methodological limitation, the pattern – where identified – supports Fishman's [32] concept of 'interactional work,' suggesting that women invest more articulatory effort to ensure communicative success. Bluntly put, while men can 'afford' to be relaxed without losing authority, women may prioritize clarity and exactness [33] to guarantee they are understood. Furthermore, the prevalence of relaxed settings in males may reflect a 'covert prestige' associated with linguistic nonchalance in Swiss German masculine identity. This 'relaxed' posture may serve as a local phonetic marker of authenticity and effortless belonging within the male speech community.

Regarding voice type, results diverge significantly from recent sociophonetic trends. While Anglophone research (e.g., [6, 34]) highlights prevalent creaky voice, particularly among young women, this study found non-modal phonation was marginal. Contrary to hypotheses derived from these studies and Peters' [12] findings on German creaky voice appeared in only 12 speakers (<10%) with no gender imbalance (7 men, 5 women), while breathy ($n=1$) and whispered voices were virtually absent. This implies a strong Swiss German preference for modal voice. Resistance may stem from negative evaluations; Anderson et al. [35] link vocal fry to sounding less competent, and Gobl and Chasaide [36] associate it with boredom, contradicting the Swiss imperative of 'friendliness' described by Rash [37]. Although lexical anglicisms are common [38], this phonetic feature has not permeated.

4.3. Geographic Distribution

Geographic analysis confirms that vocal profiles are spatially structured, largely validating the third hypothesis regarding Northeastern nasality. This finding likely results from the dorsal /r/ prevalent in the Northeast. As Graf [39] notes, uvular articulation mechanically lowers the velum to facilitate vibration or constriction against the tongue root; this creates a velopharyngeal opening that adds a 'nasal coat' to surrounding vowels. Consequently, the zone of nasality identified in this study overlaps almost perfectly with the distribution of the uvular /r/ documented in the *Dialäktatlas* [1]. This suggests that the nasal quality of Northeastern dialects may not merely be a habitual phonation choice, but an articulatory consequence of the regional consonant inventory.

Linguistically, the pharyngeal expansion observed in Central Switzerland may correspond to the morphological preservation

of full vowels in unstressed syllables – a trait distinguishing these dialects from High Alemannic schwa-reducing varieties [1]. The articulation of such open vowels (e.g., [tanə] vs. [tanə] for 'fir tree'; [viər] vs. [viər] 'four') necessitates a wider pharyngeal cavity, reinforcing the perceptual impression of 'hyperarticulation' or 'speaking to an audience.' However, this interpretation demands significant caution due to intertwined demographic and methodological constraints. First, the geographic pattern is likely skewed by demographics. Because pharyngeal expansion is already more common in women (see Section 3.2), the fact that our Central Swiss sample has slightly more female speakers naturally leads to higher rates of expansion in this region. Second, and perhaps more critically, the IRR for the pharyngeal setting was notably low ($\kappa=0.094$). This substantial disagreement indicates that while the physiological link between open vowels and pharyngeal space is theoretically plausible, its perceptual verification in this study remains tentative.

5. Conclusion

This study analyzed spontaneous speech from 127 speakers to address the lack of research on Swiss German vocal profiles. Results indicated slight to fair IRR ($\kappa<0.40$), though salient features like nasality achieved fair agreement. Socially, a gender dichotomy emerged: women exhibited pharyngeal expansion ('hyperarticulation'), while men tended toward a relaxed vocal tract ('hypoarticulation'). Non-modal phonation like creaky voice was marginal. The dominance of modal voice contrasts with the frequent borrowing of English vocabulary. This implies that while the lexicon is open to change, the vocal profile remains conservative, rejecting foreign phonetic features that may not align with local communicative norms. Geographically, distinct regional qualities were confirmed, notably a Northeastern 'nasality belt' linked to the dorsal /r/. However, the hyperarticulation observed in Central Switzerland requires cautious interpretation, as it is likely confounded by the regional overrepresentation of female speakers and compromised by the low reliability of the pharyngeal setting.

Beyond these specific interpretative challenges, the validity of the maps shown is defined by the structural constraints of the dataset. First, the sampling protocol – restricted to a single speaker per survey site – constrains local representativeness, necessitating the use of scalar interpolation to visualize broader regional trends. Second, the reliance on audio-only data impeded the assessment of visual settings, resulting in lower reliability for mandibular and labial configurations. Third, the perceptual baseline was inevitably filtered through the coders' own linguistic backgrounds (High Alemannic). Finally, the interview context itself may have induced speech accommodation [40], potentially influencing speakers to adapt features [41] to the interviewer.

Future research should validate these findings by expanding the analysis to include the full cohort of four younger speakers per location in the *Dialäktatlas* corpus. To mitigate the subjectivity of perceptual coding, future studies should triangulate these results with acoustic measurements, such as formant analysis for vocal tract settings [42] or automated algorithms for detecting creak [43]. Finally, perception experiments could investigate whether specific profiles contribute to the known social stigmatization of these dialects.

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7. Generative AI Use Disclosure

In accordance with the declaration of independent work, the large language model *Gemini* was utilized for proofreading and linguistic editing, and not for conception, structuring, data analysis, or literature synthesis.

8. References

- [1] A. Leemann et al., *Dialäktatlas – 1950 bis heute*. Zurich, Switzerland: vdf Hochschulverlag, 2025.
- [2] H. Baumgartner et al., *Sprachatlas der deutschen Schweiz*. Bern, Switzerland: Francke, 1962.
- [3] J. Laver, *The Phonetic Description of Voice Quality*. Cambridge, U.K.: Cambridge University Press, 1980.
- [4] D. Abercrombie, *Elements of General Phonetics*. Edinburgh, U.K.: Edinburgh University Press, 1967.
- [5] J. Stuart-Smith, “Glasgow: Accent and voice quality,” in *Urban Voices: Accent Studies in the British Isles*, P. Foulkes and G. J. Docherty, Eds. London, U.K.: Arnold, 1999, pp. 203–222.
- [6] H. White, J. Penney, and F. Cox, “Change in the prevalence of creaky voice over time in Australian English,” *J. Acoust. Soc. Am.*, vol. 5, Art. no. 095201, 2025, doi: 10.1121/10.0039080
- [7] E. Pépiot and A. Arnold, “Différences acoustiques inter-genres chez des bilingues Anglais/Français: Une étude des formants vocaliques et de la qualité de voix,” in *Proc. Journées d’Études sur la Parole (JEP)*, 2020, pp. 480–488.
- [8] J. B. Alonso, M. A. Ferrer, J. de León, and C. M. Travieso, “Cuantificación de la calidad de la voz para su evaluación clínica por medio del análisis acústico,” in *Proc. of the IV Jornadas en Tecnología del Habla*, 2006, pp. 203–208.
- [9] E. San Segundo and J. A. Mompean, “A simplified vocal profile analysis protocol for the assessment of voice quality and speaker similarity,” *J. Voice*, vol. 31, no. 5, pp. 644.e11–644.e27, 2017, doi: 10.1016/j.jvoice.2017.01.005
- [10] J. Beck and F. Schaeffler, “Voice quality variation in Scottish adolescents: Gender versus geography,” in *Proc. 18th Int. Congr. Phonetic Sciences (ICPhS)*, Glasgow, U.K., 2015.
- [11] E. Mendoza, N. Valencia, J. Muñoz, and H. Trujillo, “Differences in voice quality between men and women: Use of the long-term average spectrum (LTAS),” *J. Voice*, vol. 10, no. 1, pp. 59–66, 1996.
- [12] J. Peters, “Regionale Variation der Stimmqualität: Eine Pilotstudie zum hoch- und niederdeutschen im Bersenbrücker Land,” *Zeitschrift für Dialektologie und Linguistik*, vol. 85, no. 1, pp. 1–34, 2018.
- [13] J. Vetsch, *Die Laute der Appenzeller Mundarten*. Frauenfeld, Switzerland: Huber & Co., 1910.
- [14] W. Wiget, *Die Laute der Toggenburger Mundarten*. Frauenfeld, Switzerland: Huber & Co., 1916.
- [15] J. Berger, *Die Laute der Mundarten des St. Galler Rheintals und der angrenzenden vorarlbergischen Gebiete*. Frauenfeld, Switzerland: Huber & Co., 1913.
- [16] C. Streiff, *Die Laute der Glarner Mundarten*. Frauenfeld, Switzerland: Huber & Co., 1915.
- [17] A. Weber, *Die Mundart des Zürcher Oberlandes*. Frauenfeld, Switzerland: Huber & Co., 1923.
- [18] K. Schmid, *Die Mundart des Amtes Entlebuch im Kanton Luzern*. Frauenfeld, Switzerland: Huber & Co., 1915.
- [19] W. Clauß, *Die Mundart von Uri: Laut- und Flexionslehre*. Frauenfeld, Switzerland: Huber & Co., 1929.
- [20] R. T. Sataloff and S. Linville, “The effect of age on the voice,” in *Vocal Health and Pedagogy: Science, Assessment, and Treatment*, 2005, pp. 319–338.
- [21] A. Leemann, P. Jeszenszky, C. Steiner, M. Studerus, and J. Messerli, “Linguistic fieldwork in a pandemic: Supervised data collection combining smartphone recordings and videoconferencing,” *Linguist. Vanguard*, vol. 6, no. S3, Art. no. 20200061, 2020, doi: 10.1515/lingvan-2020-0061
- [22] R Core Team, *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria, 2026. [Online]. Available: <https://www.R-project.org/> (Accessed: Jan. 09, 2026).
- [23] J. R. Landis and G. G. Koch, “An application of hierarchical kappa-type statistics in the assessment of majority agreement among multiple observers,” *Biometrics*, vol. 33, no. 2, pp. 363–374, 1977.
- [24] A. Kassambara, *Inter-Rater Reliability Essentials: Practical Guide in R*, 1st ed., 2019.
- [25] QGIS Development Team, *QGIS Geographic Information System*, QGIS Association, 2025. [Online]. doi: 10.5281/zenodo.6139224 (Accessed: Jan. 15, 2026).
- [26] T. Cover and P. Hart, “Nearest neighbor pattern classification,” *IEEE Trans. Inf. Theory*, vol. 13, no. 1, pp. 21–27, 1967.
- [27] A. L. Webb, P. N. Carding, I. J. Deary, K. MacKenzie, N. Steen, and J. A. Wilson, “The reliability of three perceptual evaluation scales for dysphonia,” *Eur. Arch. Otorhinolaryngol.*, vol. 261, no. 8, pp. 429–434, 2004, doi: 10.1007/s00405-003-0707-7
- [28] C. Llamas, D. Watt, and A. E. MacFarlane, “Estimating the relative sociolinguistic salience of segmental variables in a dialect boundary zone,” *Front. Psychol.*, vol. 7, Art. no. 1163, 2016, doi: 10.3389/fpsyg.2016.01163
- [29] A. Guex and M. Pithon, *Manuel de phonétique française*. Lausanne, Switzerland: École de français moderne de l’Université de Lausanne, 1975.
- [30] J. Beck, *Vocal Profile Analysis Scheme: A User’s Manual*. Edinburgh, U.K.: Queen Margaret Univ. College, Speech Science Centre, 2007.
- [31] E. San Segundo, P. Foulkes, P. French, P. Harrison, V. Hughes, and C. Kavanagh, “The use of the vocal profile analysis for speaker characterization: Methodological proposals,” *J. Int. Phonetic Assoc.*, vol. 49, no. 3, pp. 353–380, 2019, doi: 10.1017/S0025100318000130
- [32] P. M. Fishman, “The work women do,” *Soc. Probl.*, vol. 25, no. 4, pp. 397–406, 1978.
- [33] R. Lakoff, “Language and woman’s place,” *Lang. Soc.*, vol. 2, no. 1, pp. 45–80, 1973.
- [34] E. Gold, C. Kirchhübel, K. Earnshaw, and S. Ross, “Regional variation in British English voice quality,” *English World-Wide*, vol. 43, no. 1, pp. 96–123, 2022, doi: 10.1075/eww.20007.gol
- [35] R. C. Anderson, C. A. Klostad, W. J. Mayew, and M. Venkatachalam, “Vocal fry may undermine the success of young women in the labor market,” *PLoS ONE*, vol. 9, no. 5, Art. no. e97506, 2014, doi: 10.1371/journal.pone.0097506
- [36] C. Gobl and A. Ni Chasaide, “The role of voice quality in communicating emotion, mood and attitude,” *Speech Commun.*, vol. 40, 1–2, pp. 189–212, 2003.
- [37] F. Rash, “Linguistic politeness and greeting rituals in German-speaking Switzerland,” *Linguistik Online*, vol. 20, no. 3, pp. 47–72, 2004, doi: 10.13092/lo.20.1063
- [38] S. Krome, “Skypen, faken, toppen und liken: Anglizismen im Deutschen als Indikatoren gesellschaftlichen und orthografischen Wandels,” *Muttersprache*, vol. 128, no. 2, pp. 105–122, 2018.

- [39] M. H. Graf, *Thurgauer Mundart in Geschichte und Gegenwart*. Bern, Switzerland: Schweizerische Akademie der Geistes- und Sozialwissenschaften, 2014.
- [40] H. Giles and P. Powesland, "Accommodation theory," in *Sociolinguistics: A Reader*, N. Coupland and A. Jaworski, Eds. London, U.K.: Macmillan Education, 1997, pp. 232–239.
- [41] H. White, J. Penney, A. Gibson, A. Szakay, and F. Cox, "Convergence of creaky voice use in Australian English," in *Proc. 20th Int. Congr. Phonetic Sciences (ICPhS)*, 2023, pp. 1791–1795.
- [42] M. Bauer, "Articulatory conflict and laryngeal height," in *Proc. Int. Congr. Phonetic Sciences (ICPhS)*, 2011, pp. 292–295.
- [43] H. White, J. Penney, A. Gibson, A. Szakay, and F. Cox, "Evaluating automatic creaky voice detection methods," *J. Acoust. Soc. Am.*, vol. 152, no. 3, pp. 1476–1486, 2022. doi: 10.1121/10.0013888